

Data Warehouse, Data Mart, Data Lake & Data Lakehouse

1. Data Warehouse

A Data Warehouse is a centralized repository that stores structured, cleaned, and processed data collected from multiple sources for reporting, analytics, and business intelligence purposes.

Key Characteristics:

- Stores structured data only
- Schema-on-write (data is cleaned before storing)
- Used for historical analysis
- High data consistency and accuracy

Real-Life Example:

A bank uses a data warehouse to store customer transactions, loan data, and account details to generate monthly financial reports.

2. Data Mart

A Data Mart is a smaller, subject-specific subset of a data warehouse designed for a particular department or business function.

Key Characteristics:

- Focused on a single business area
- Faster access for teams
- Lower storage size

Real-Life Example:

The HR department accesses a data mart containing employee performance and payroll data.

3. Data Lake

A Data Lake is a centralized repository that stores raw data in its original format, including structured, semi-structured, and unstructured data.

Key Characteristics:

- Stores raw data
- Schema-on-read
- Supports big data and machine learning

Real-Life Example:

Netflix stores user clicks, watch history, and logs in a data lake for recommendation systems.

4. Data Lakehouse

A Data Lakehouse combines the best features of data lakes and data warehouses by enabling structured analytics on raw data.

Key Characteristics:

- Supports both BI and ML
- ACID transactions
- Lower cost than data warehouse

Real-Life Example:

Databricks Lakehouse used by companies for analytics and AI workloads.

5. Comparison Table

Feature	Data Warehouse	Data Mart	Data Lake	Data Lakehouse
Data Type	Structured	Structured	All types	All types
Schema	Schema-on-write	Schema-on-write	Schema-on-read	Hybrid
Use Case	BI & Reporting	Department Analysis	Big Data & ML	BI + ML

Feature	Data Warehouse	Data Mart	Data Lake	Data Lakehouse
Cost	High	Medium	Low	Medium
Users	Business Analysts	Teams	Data Scientists	All users

6. Simple Visualization (Text Diagram)

Data Sources → Data Lake (Raw Data) → Data Lakehouse (Analytics + ML) → Data Warehouse → Data Mart → Business Users

7. Interview Summary

- Use Data Warehouse for clean reporting
- Use Data Lake for raw big data and ML
- Use Data Lakehouse for modern analytics
- Use Data Mart for department-specific needs